Physics 9th – Chapter-7-Density and Temperature (Short Response Questions)

Q1. Two liquids A and B have densities 1 g mL^{-1} and 1.2 g mL^{-1}. Which liquid is on top, and why?

Ans:

- Liquid A floats on top of liquid B.
- Density is mass per unit volume; liquid A (1 g/mL) is less dense than liquid B (1.2 g/mL).
- Liquid B has more matter in the same volume, so it settles at the bottom and A stays above it.

Q2. Write a method to find the volume and density of a human body.

Ans:

- Volume (displacement method): submerge the body in a water-filled container; volume of body = final volume (Vf) − initial volume (Vi) of water.
- Mass: measure the mass of the person with a balance (weighing scale).
- Density = Mass of body / Volume of body ($\rho = m/V$), in kg/m^3.

Q3. How is plasma the fourth state of matter? Give a reason.

Ans:

- Plasma consists of positive ions, free electrons and neutral atoms in gaseous state; it exists at very high temperature or pressure, or by using a high electric and magnetic field.
- On continuous heating, K.E. of gas molecules increases and they collide powerfully; electrons are removed and atoms become positive ions. This ionic state is called plasma.
- It is hot, bright, highly ionized and conducts current (free electrons and ions). 99% of the visible universe (Sun, stars, aurora, neon light) is plasma, so it is called the fourth state of matter.

Q4. Why is water not used in liquid-in-glass thermometers?

Ans:

- Mercury and alcohol expand more than liquids like water when heated, so thermometers having them are more sensitive; water expands less.
- Mercury is used because of its smaller melting point and higher boiling point; water has a narrow liquid range (0°C to 100°C), so its range would be limited.
- A thermometric liquid should expand uniformly with temperature, as mercury does.

Q5. Can we increase internal energy of a substance without increasing its temperature?

Ans:

- Yes. Internal energy is the total kinetic + potential energy of particles, while temperature depends only on their average K.E.
- At melting and boiling points, the heat given only increases the gap between molecules by decreasing intermolecular forces (potential energy) to change the state.
- It is not used to increase K.E., so internal energy increases but temperature stays the same.

Q6. Why are fixed point scales required for thermometers? What difficulties are there when setting fixed points?

Ans:

- Without a calibrated scale a thermometer is valueless. We need two fixed (reference) points, lower and upper, which should be accessible and reproducible; the gap between them is divided into equal divisions.
- Lower point (melting ice): 0°C, 32°F, 273 K. Upper point (boiling water): 100°C, 212°F, 373 K.
- Difficulty: a standard degree of hotness/coldness must be accessible and reproducible, and scales differ in values and divisions (100 on Celsius and Kelvin, 180 on Fahrenheit), so these must be decided first.

Q7. Mercury is replaced with alcohol in liquid-in-glass thermometers. Discuss the possible change in sensitivity and range.

Ans:

- Sensitivity: alcohol expands more than mercury and water when heated, so smaller temperature changes give larger rise of the liquid; sensitivity increases.
- Range: mercury is used because of its smaller melting point and higher boiling point; a liquid with a narrower boiling point range gives a narrower range.
- Alcohol boils at a lower temperature, so the range decreases (high temperatures cannot be measured).

Q8. Why is −273.15°C called absolute zero? Can we achieve this temperature?

Ans:

- On removing heat, particles keep slowing down and losing K.E. till they no longer move and have least or no K.E.; they cannot collide, so pressure is zero (P = 0 Pa).
- This temperature is called absolute zero: “the lowest possible temperature of a substance at which its particles have least kinetic energy”.
- Its value is 0 K = −273.15°C. It is only a theoretical limit and cannot practically be achieved.

Q9. Why is a thermocouple thermometer suitable to measure high temperatures but not a liquid-in-glass thermometer?

Ans:

- Thermocouple has two wires of different metals joined at a hot junction; heating it produces a voltage difference (linearly proportional to temperature) measured by a voltmeter.
- It has no liquid or glass bulb, so it is used for furnaces, kilns, engines and other industrial equipment.
- Liquid-in-glass thermometer depends on the expansion of mercury/alcohol in a glass tube, so its range is limited by the liquid; it is used for moderate temperatures (homes, schools, labs).

Q10. Can we increase the sensitivity of a liquid-in-glass thermometer without changing its range?

Ans:

- Yes, by using a capillary tube of smaller (narrower) diameter: it has a smaller volume of liquid, which reacts quickly to heat and rises quickly, detecting smaller changes.

- Also by using a liquid like mercury or alcohol, which expands more than liquids like water.
- Range stays unchanged because it depends on the bulb size and the melting/boiling points of the liquid, not on the tube diameter.

Q11. One student claims to have constructed a more sensitive liquid-in-glass thermometer. How can her claim be verified?

Ans:

- Place her thermometer and a standard one in a glass of water at room temperature and let them settle.
- Add a few drops of hot water: the thermometer whose liquid rises quickly and shows the smaller change in temperature is more sensitive (e.g., 0.1°C least count detects smaller changes than 1°C).
- Also check that her tube has a smaller diameter or a more expansive liquid than the standard one.